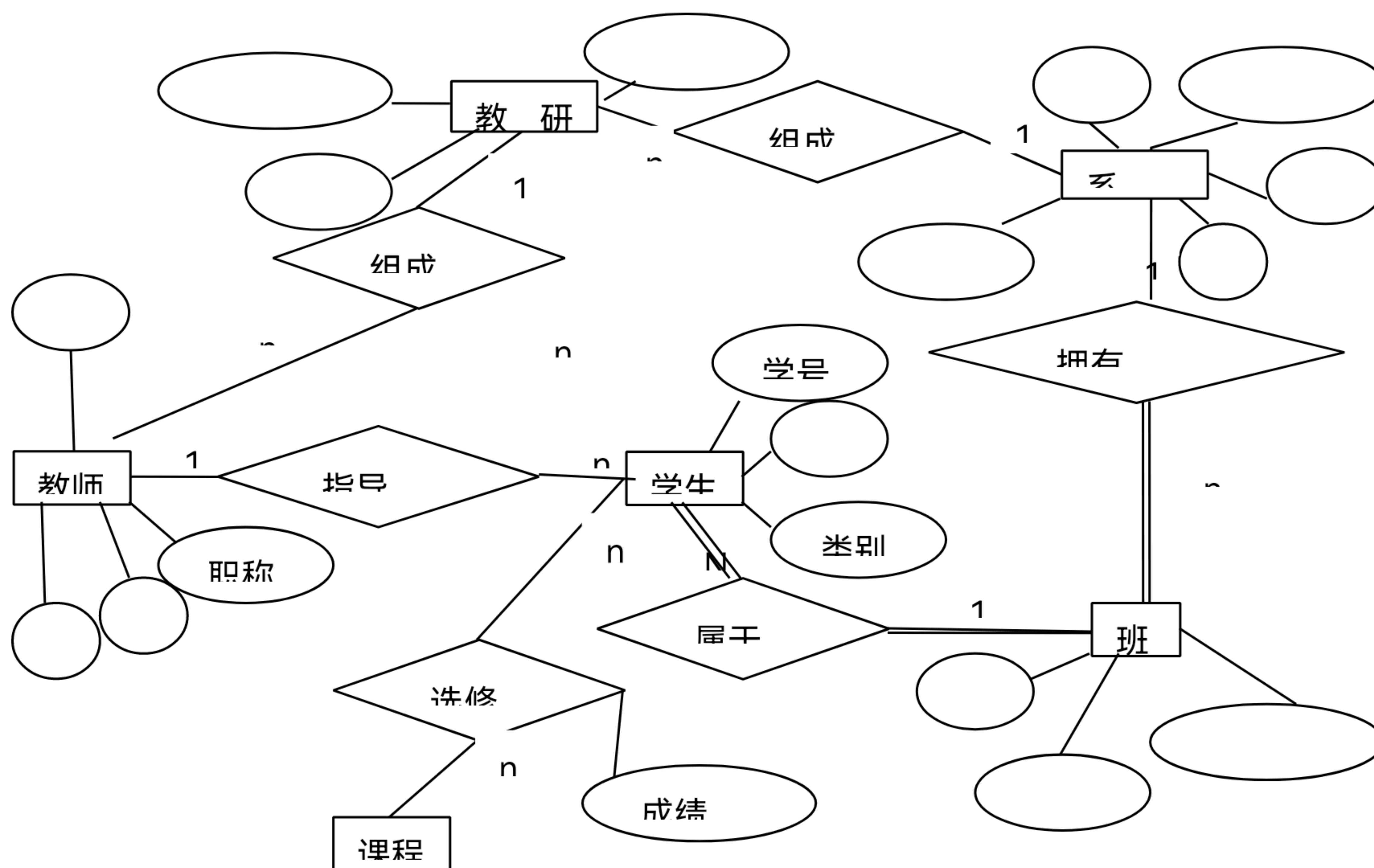


Assignment of Database Design:

1. 一个学校有若干个系,每个系有若干个教研室和若干个班级,每个班级有若干个学生,每个教研室有若干个教师,其中有的教授和副教授,指导若干个研究生.每个学生可以选多门课,每门课程有多个学生选修,每个学生选修了一门课,经过考试,就有一个成绩.请画出该学校管理系统的E-R图。

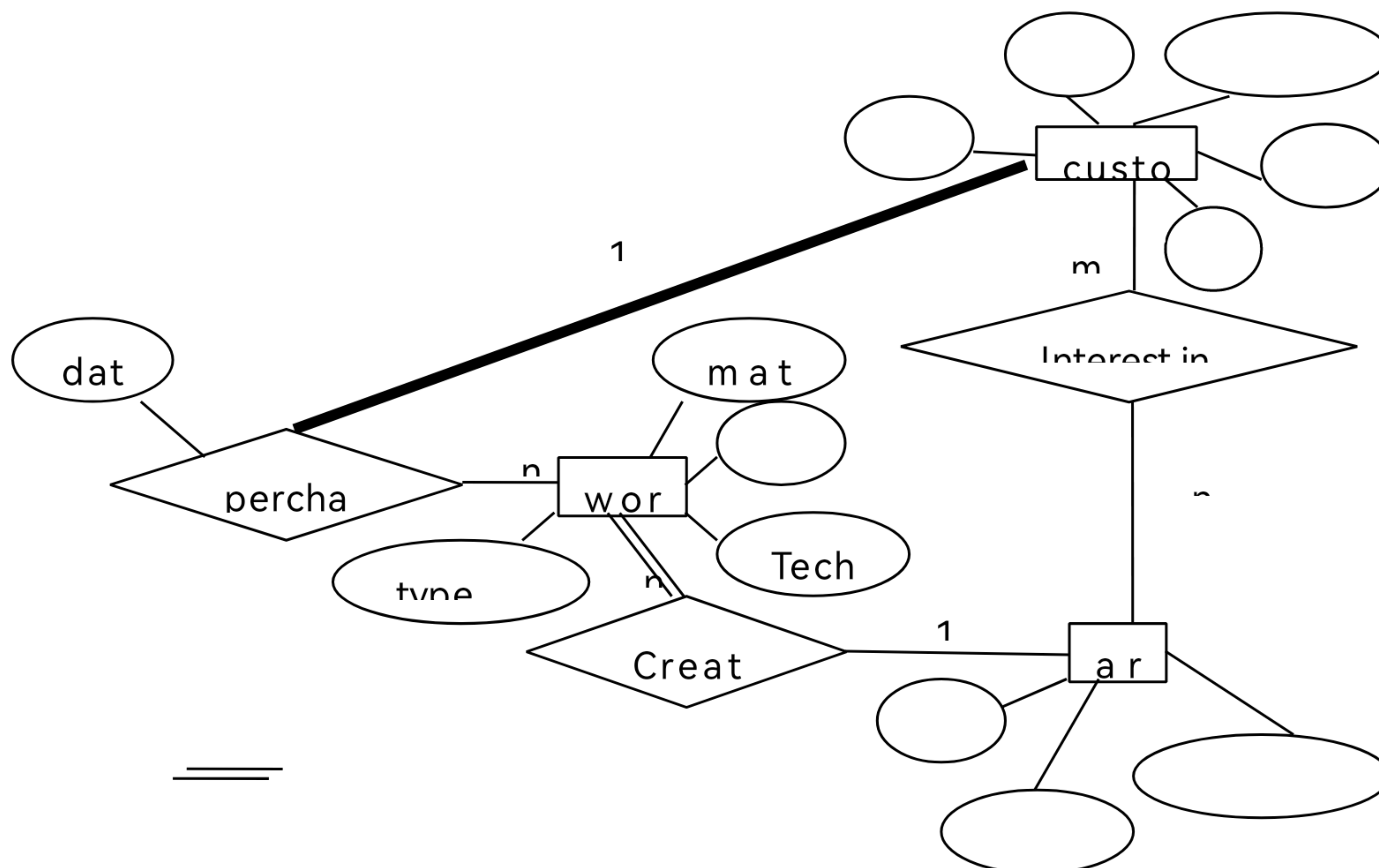


2 . Represent, by means of one or more relations, the information about **art gallery**, including the **artists**, their **works** and the **customers**. Artists are described by **their name, birthplace, age, and style of art**. The gallery keeps information about artists regardless of whether the museum actually exhibits or has previously sold any artwork produced by the artist.

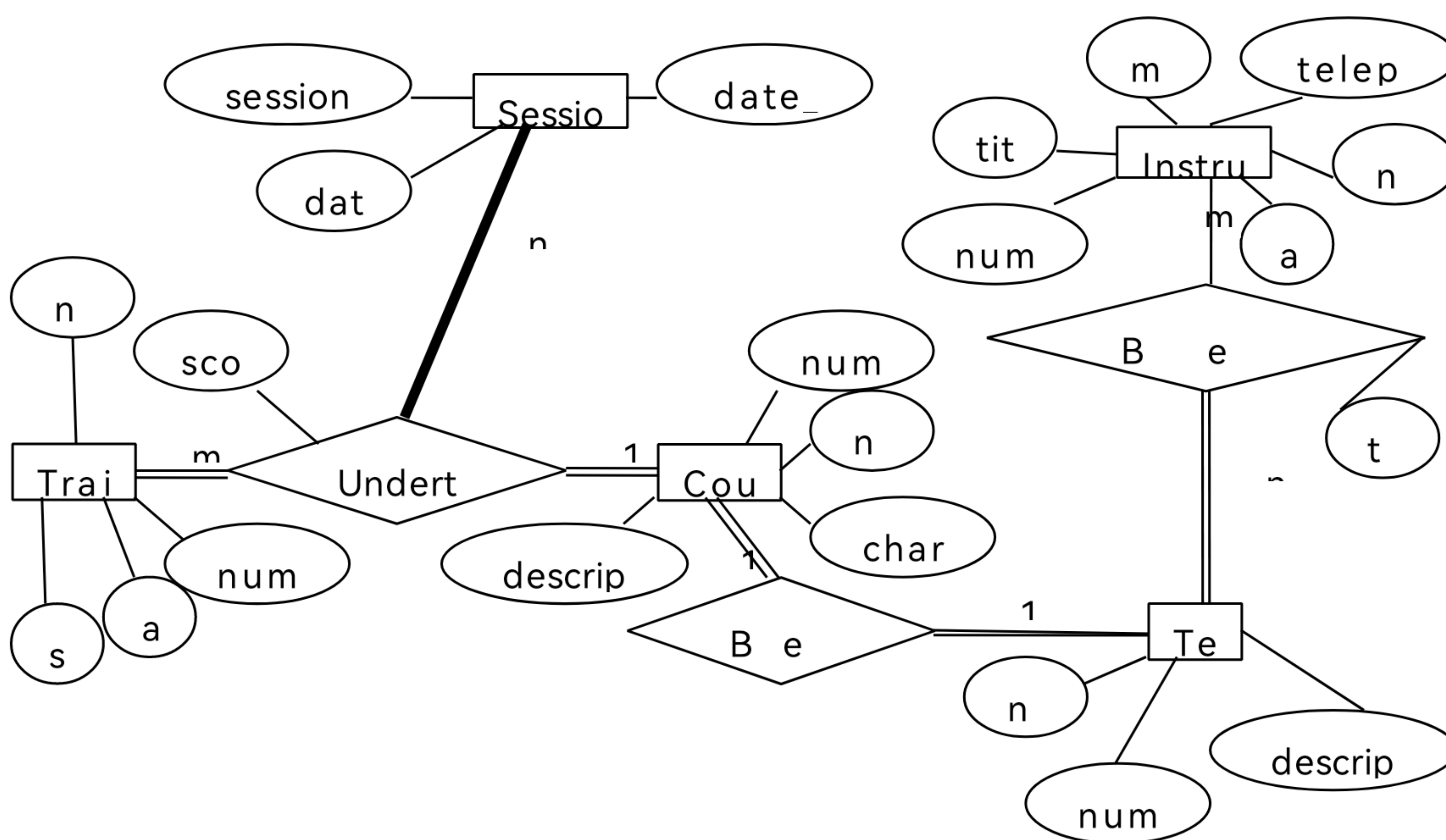
For each piece of artwork exhibited in or previously sold by the gallery, **the artist, the year it was made, its title, and its price must be stored**.

The museum collects only artworks which are either paintings or sculptures. For paintings, the painting technique must be recorded and for sculptures, the material used must be stored.

The gallery also keeps track of purchases of art works by customers and the interest of customers in particular artists. **A customer recorded may not have indicated his or her interest in particular artists**, but must have purchased at least one art work.



3. Map the following E-R diagram to logical schemas



4. (*point: 12*) The table shown in the following relation contains purchase information of a supermarket, including the customer information such as number, name, address(*Cust_Number*, *cust_name*, *Adrress*), the order information such as order number, order date, date due, total price(*Ord_number*, *Ord-date*, *Date_due*, *Total_price*), and the product information such as the product number, product name, price, quantity(*P_number*, *P-name*, *type*, *price*, *quantity*). One customer can play one or more orders and each order is played by one customer. Each order

includes one or more products, one product can be contained in one or more orders.

Purchase(Cust_Number, cust_name, Address, Ord_number, Ord-date, Date_due, Total_price, P_number, P-name, type, price, quantity)

- (a) Identify all the functional dependencies within the above relation; (point: 2)
- (b) Identify the primary key of this table; (point:2)
- (c) What normal form is the relation in? Explain your answer.(point: 2)
- (d) Apply normalization until the relations are BCNF. State the reasons behind each decomposition. (point:6)